Publications

Yinghao Xu (Justin) · HKUST

Total: 60 publications · * Equal contribution · + Corresponding author

2026

1. **Causal World Modeling for Robot Control**
Lin Li*, Qihang Zhang*, Yiming Luo*, Shuai Yang (RobbyAnt), Ruilin Wang, Fei Han, Mingrui Yu, Zelin Gao, Nan Xue, Xing Zhu, Yujun Shen, **Yinghao Xu**
arXiv 2026

2. **Advancing Open-source World Models**
Zelin Gao, Qiuyu Wang, Yanhong Zeng, Jiapeng Zhu, Ka Leong Cheng, Yixuan Li, Hanlin Wang, **Yinghao Xu**, Shuailei Ma, Yihang Chen, Jie Liu, Yansong Cheng, Yao Yao, Jiayi Zhu, Yihao Meng, Kecheng Zheng, Qingyan Bai, Jingye Chen, Zehong Shen, Yue Yu, Xing Zhu, Yujun Shen, Hao Ouyang
arXiv 2026

3. **Masked Depth Modeling for Spatial Perception**
Bin Tan, Changjiang Sun, Xiage Qin, Hanat Adai, Zelin Fu, Tianxiang Zhou, Han Zhang, **Yinghao Xu**, Xing Zhu, Yujun Shen, Nan Xue
arXiv 2026

4. **SceneScribe-1M: A Large-Scale Video Dataset with Comprehensive Geometric and Semantic Annotations** *[Highlight]*
Yunnan Wang, Kecheng Zheng, Jianyuan Wang, Minghao Chen, David Novotny, Christian Rupprecht, **Yinghao Xu**, Xing Zhu, Wenjun Zeng, Xin Jin, Yujun Shen
CVPR 2026

5. **Ditto: Scaling Instruction-Based Video Editing with a High-Quality Synthetic Dataset** *[Highlight]*
Qingyan Bai, Qiuyu Wang, Hao Ouyang, Yue Yu, Hanlin Wang, Wen Wang, Ka Leong Cheng, Shuailei Ma, Yanhong Zeng, Zichen Liu, **Yinghao Xu**, Yujun Shen, Qifeng Chen
CVPR 2026

6. **Geometric Context Transformer for Streaming 3D Reconstruction**
Linzhuo Chen*, Jian Gao*, Yihang Chen, Ka Leong Cheng, Yipengjing Sun, Liangxiao Hu, Nan Xue, Xing Zhu, Yujun Shen, Yao Yao+, **Yinghao Xu+**
arXiv 2026

7. **Mixture of Contexts for Long Video Generation**
Shengqu Cai, Ceyuan Yang, Lvmin Zhang, Yuwei Guo, Junfei Xiao, Ziyan Yang, **Yinghao Xu**, Zhenheng Yang, Alan Yuille, Leonidas Guibas, Maneesh Agrawala, Lu Jiang, Gordon Wetzstein
ICLR 2026

2025

8. **FLARE: Feed-forward Geometry, Appearance and Camera Estimation from Uncalibrated Sparse Views**

Shangzhan Zhang*, Jianyuan Wang*, **Yinghao Xu***, Nan Xue, Christian Rupprecht, Xiaowei Zhou, Yujun Shen, Gordon Wetzstein

CVPR 2025

9. **Video World Models with Long-term Spatial Memory**

Tong Wu, Shuai Yang, Ryan Po, **Yinghao Xu**, Ziwei Liu, Dahua Lin, Gordon Wetzstein

NeurIPS 2025

10. **Interspatial Attention for Efficient 4D Human Video Generation**

Ruizhi Shao*, **Yinghao Xu***, Yujun Shen, Ceyuan Yang, Yang Zheng, Changan Chen, Yebin Liu, Gordon Wetzstein

SIGGRAPH 2025

11. **GroomLight: Hybrid Inverse Rendering for Relightable Human Hair Appearance Modeling**

Yang Zheng, Menglei Chai, Delio Vicini, Yuxiao Zhou, **Yinghao Xu**, Leonidas Guibas, Gordon Wetzstein, Thabo Beeler

CVPR 2025

12. **CameraCtrl II: Dynamic Scene Exploration via Camera-controlled Video Diffusion Models**

Hao He, Ceyuan Yang, Shanchuan Lin, **Yinghao Xu**, Meng Wei, Liangke Gui, Qi Zhao, Gordon Wetzstein, Lu Jiang, Hongsheng Li

ICCV 2025

13. **Edicho: Consistent Image Editing in the Wild**

Qingyan Bai, Hao Ouyang, **Yinghao Xu**, Qiuyu Wang, Ceyuan Yang, Ka Leong Cheng, Yujun Shen, Qifeng Chen

ICCV 2025

14. **3DitScene: Editing Any Scene via Language-guided Disentangled Gaussian Splatting**

Qihang Zhang, **Yinghao Xu**, Chaoyang Wang, Hsin-Ying Lee, Gordon Wetzstein, Bolei Zhou, Ceyuan Yang

ICLR 2025

15. **CameraCtrl: Enabling Camera Control for Video Diffusion Models**

Hao He, **Yinghao Xu**, Yuwei Guo, Gordon Wetzstein, Bo Dai, Hongsheng Li, Ceyuan Yang

ICLR 2025

2024

16. **Flow as the Cross-domain Manipulation Interface** *[Oral at X-Embodiment Workshop]*

Mengda Xu, Zhenjia Xu, **Yinghao Xu**, Cheng Chi, Gordon Wetzstein, Manuela Veloso, Shuran Song

CoRL 2024

17. **Collaborative Video Diffusion: Consistent Multi-video Generation with Camera Control**

Zhengfei Kuang*, Shengqu Cai*, Hao He, **Yinghao Xu**, Hongsheng Li, Leonidas Guibas, Gordon Wetzstein

NeurIPS 2024

18. **FiVA: Fine-grained Visual Attribute Dataset for Text-to-Image Diffusion Models**

Tong Wu, **Yinghao Xu**, Ryan Po, Mengchen Zhang, Guandao Yang, Jiaqi Wang, Ziwei Liu, Dahua Lin, Gordon Wetzstein

NeurIPS 2024 Dataset Benchmark

19. **Representing Long Volumetric Video with Temporal Gaussian Hierarchy**

Zhen Xu*, **Yinghao Xu***, Zhiyuan Yu*, Sida Peng, Jiaming Sun, Hujun Bao, Xiaowei Zhou

SIGGRAPH ASIA 2024 (Journal Track)

20. **Real-time 3D-aware Portrait Editing from a Single Image**

Qingyan Bai, Zifan Shi, **Yinghao Xu**, Hao Ouyang, Qiuyu Wang, Ceyuan Yang, Xuan Wang, Gordon Wetzstein, Yujun Shen, Qifeng Chen

ECCV 2024

21. **GRM: Large Gaussian Reconstruction Model for Efficient 3D Reconstruction and Generation**

Yinghao Xu*, Zifan Shi*, Wang Yifan, Hansheng Chen, Ceyuan Yang, Sida Peng, Yujun Shen, Gordon Wetzstein

ECCV 2024

22. **BerfScene: Bev-conditioned Equivariant Radiance Fields for Infinite 3D Scene Generation**

Qihang Zhang, **Yinghao Xu**, Yujun Shen, Bo Dai, Bolei Zhou, Ceyuan Yang

CVPR 2024

23. **SceneWiz3D: Towards Text-guided 3D Scene Composition**

Qihang Zhang, Chaoyang Wang, Aliaksandr Siarohin, Peiye Zhuang, **Yinghao Xu**, Ceyuan Yang, Dahua Lin, Bolei Zhou, Sergey Tulyakov, Hsin-Ying Lee

CVPR 2024

24. **Gaussian Shell Maps for Efficient 3D Human Generation**

Rameen Abdal*, Wang Yifan*, Zifan Shi*, **Yinghao Xu**, Ryan Po, Zhengfei Kuang, Qifeng Chen, Dit-Yan Yeung, Gordon Wetzstein

CVPR 2024

25. **PF-LRM: Pose-Free Large Reconstruction Model for Joint Pose and Shape Prediction** *[Spotlight]*

Peng Wang, Hao Tan, Sai Bi, **Yinghao Xu**, Fujun Luan, Kalyan Sunkavalli, Wenping Wang, Zexiang Xu, Kai Zhang

ICLR 2024

26. **DMV3D:Denoising Multi-View Diffusion using 3D Large Reconstruction Model** *[Spotlight]*

Yinghao Xu, Hao Tan, Fujun Luan, Sai Bi, Peng Wang, Jiahao Li, Zifan Shi, Kalyan Sunkavalli, Gordon Wetzstein, Zexiang Xu, Kai Zhang

ICLR 2024

27. **Instant3D: Fast Text-to-3D with Sparse-View Generation and Large Reconstruction Model**

Jiahao Li, Hao Tan, Kai Zhang, Zexiang Xu, Fujun Luan, **Yinghao Xu**, Yicong Hong, Kalyan Sunkavalli, Greg Shakhnarovich, Sai Bi

ICLR 2024

28. **Efficient 3D Articulated Human Generation with Layered Surface Volumes** *[Spotlight]*
Yinghao Xu, Wang Yifan, Alexander W. Bergman, Menglei Chai, Bolei Zhou, Gordon Wetzstein
3DV 2024

2023

29. **Learning Modulated Transformation in GANs**
Ceyuan Yang, Qihang Zhang, **Yinghao Xu**, Jiapeng Zhu, Yujun Shen, Bo Dai
NeurIPS 2023

30. **In-Domain GAN Inversion for Real Image Editing**
Jiapeng Zhu*, Yujun Shen*, **Yinghao Xu**, Deli Zhao, Qifeng Chen, Bolei Zhou
TPAMI 2023

31. **Benchmarking and Analyzing 3D-aware Image Synthesis with a Modularized Codebase**
Qiuyu Wang, Zifan Shi, Kecheng Zheng, **Yinghao Xu**, Sida Peng, Yujun Shen+
NeurIPS 2023

32. **One-Shot Generative Domain Adaptation**
Ceyuan Yang*, Yujun Shen*, Zhiyi Zhang, **Yinghao Xu**, Jiapeng Zhu, Zhirong Wu, Bolei Zhou
ICCV 2023

33. **Learning 3D-aware Image Synthesis with Unknown Pose Distribution**
Zifan Shi*, Yujun Shen*, **Yinghao Xu**, Sida Peng, Yiyi Liao, Sheng Guo, Qifeng Chen, Dit-Yan Yeung
CVPR 2023

34. **DiscoScene: Spatially Disentangled Generative Radiance Field for Controllable 3D-aware Scene Synthesis** *[Highlight presentation]*
Yinghao Xu, Menglei Chai, Zifan Shi, Sida Peng, Ivan Skorokhodov, Aliaksandr Siarohin, Ceyuan Yang, Yujun Shen, Hsin-Ying Lee, Bolei Zhou, Sergey Tulyakov
CVPR 2023

35. **GLeaD: Improving GANs with A Generator-Leading Task**
Qingyan Bai, Ceyuan Yang, **Yinghao Xu**, Xihui Liu, Yujiu Yang, Yujun Shen
CVPR 2023

36. **Towards Smooth Video Composition**
Qihang Zhang, Ceyuan Yang, Yujun Shen, **Yinghao Xu**, Bolei Zhou
ICLR 2023

37. **3D Generation on ImageNet** *[Oral]*
Ivan Skorokhodov, Aliaksandr Siarohin, **Yinghao Xu**, Jian Ren, Hsin-Ying Lee, Peter Wonka, Sergey Tulyakov
ICLR 2023

38. **GH-Feat: Learning Versatile Generative Hierarchical Features from GANs**
Yinghao Xu, Yujun Shen, Jiapeng Zhu, Ceyuan Yang, Bolei Zhou
TPAMI 2023

39. **Neural Body: Implicit Neural Representations with Structured Latent Codes for Novel View Synthesis of Dynamic Humans** *[Best paper candidate]*
Sida Peng, Yuanqing Zhang, **Yinghao Xu**, Qianqian Wang, Qing Shuai, Hujun Bao, Xiaowei Zhou
TPAMI 2023

2022

40. **Improving 3D-aware Image Synthesis with A Geometry-aware Discriminator** *[Spotlight]*
Zifan Shi, **Yinghao Xu**, Yujun Shen, Deli Zhao, Qifeng Chen, Dit-Yan Yeung
NeurIPS 2022

41. **Improving GANs with A Dynamic Discriminator**
Ceyuan Yang*, Yujun Shen*, **Yinghao Xu**, Deli Zhao, Bo Dai, Bolei Zhou
NeurIPS 2022

42. **High-fidelity GAN Inversion with Padding Space**
Qingyan Bai*, **Yinghao Xu***, Jiapeng Zhu, Weihao Xia, Yujiu Yang, Yujun Shen
ECCV 2022

43. **Semantic-Aware Implicit Neural Audio-Driven Video Portrait Generation** *[Oral]*
Xian Liu, **Yinghao Xu**, Qianyi Wu, Hang Zhou, Wayne Wu, Bolei Zhou
ECCV 2022

44. **Learning Hierarchical Cross-Modal Association for Co-Speech Gesture Generation**
Xian Liu, Qianyi Wu, Hang Zhou, **Yinghao Xu**, Xinyi Lin, Xiaowei Zhou, Wayne Wu, Bolei Zhou
CVPR 2022

45. **Region-Based Semantic Factorization in GANs**
Jiapeng Zhu, Yujun Shen, **Yinghao Xu**, Deli Zhao, Qifeng Chen
ICML 2022

46. **3D-aware Image Synthesis via Learning Structural and Textural Representations**
Yinghao Xu, Sida Peng, Ceyuan Yang, Yujun Shen, Bolei Zhou
CVPR 2022

47. **Improving GAN Equilibrium by Raising Spatial Awareness**
Jianyuan Wang, Ceyuan Yang, **Yinghao Xu**, Yujun Shen, Hongdong Li, Bolei Zhou
CVPR 2022

48. **Cross-Model Pseudo-Labeling for Semi-Supervised Action Recognition** *[Oral]*
Yinghao Xu, Fangyun Wei, Xiao Sun, Ceyuan Yang, Yujun Shen, Bo Dai, Bolei Zhou, Stephen Lin
CVPR 2022

2021

49. **Data-Efficient Instance Generation from Instance Discrimination**
Ceyuan Yang, Yujun Shen, **Yinghao Xu**, Bolei Zhou
NeurIPS 2021

50. **Learning Object-Compositional Neural Radiance Field for Editable Scene Rendering**
Bangbang Yang, Yinda Zhang, **Yinghao Xu**, Yijin Li, Han Zhou, Hujun Bao, Guofeng Zhang, Zhaopeng Cui
NeurIPS 2021

51. **Generative Hierarchical Features from Synthesizing Images** *[Oral]*
Yinghao Xu*, Yujun Shen*, Jiapeng Zhu, Ceyuan Yang, Bolei Zhou
CVPR 2021

52. **CompConv: A Compact Convolution Module for Efficient Feature Learning**
Chen Zhang, **Yinghao Xu**, Yujun Shen+
CVPRW 2021

53. **Decorating Your Own Bedroom: Locally Controlling Image Generation with Generative Adversarial Networks**
Chen Zhang, **Yinghao Xu**, Yujun Shen+
CVPRW 2021

2020

54. **Dense RepPoints: Representing Visual Objects with Dense Point Sets**
Yinghao Xu*, Ze Yang*, Han Xue*, Raquel Urtasun, Liwei Wang, Stephen Lin, Han Hu
ECCV 2020

55. **Temporal Pyramid Network for Action Recognition**
Ceyuan Yang*, **Yinghao Xu***, Jianping Shi, Bo Dai, Bolei Zhou
CVPR 2020

2019

56. **A Main/Subsidiary Network Framework for Simplifying Binary Networks**
Yinghao Xu, Xin Dong, Yudian Li, Hao Su
CVPR 2019

Other / Preprints

57. **Exploring Sparse MoE in GANs for Text-conditioned Image Synthesis**
Jiapeng Zhu, Ceyuan Yang, Zifan Shi, Kecheng Zheng, **Yinghao Xu**, Yujun Shen
arxiv

58. **3D Generative Models: A Survey**
Zifan Shi*, Sida Peng*, **Yinghao Xu***, Andreas Geiger, Yiyi Liao+, Yujun Shen+
arXiv

59. **Unsupervised Landmark Learning from Unpaired Data**
Yinghao Xu, Ceyuan Yang, Ziwei Liu, Bo Dai, Bolei Zhou
arxiv

60. **Video Representation Learning with Visual Tempo Consistency**
Ceyuan Yang, **Yinghao Xu**, Bo Dai, Bolei Zhou
arxiv